

2017 Ph H2 Q1

Section: Our Dynamic Universe

Topic: Constant acceleration & Doppler effect

(a) Train acceleration

(i) 0.32 m s^{-2} means the velocity increases by 0.32 m s^{-1} every second.

(ii) Distance in 25 s from rest: $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 0.32 \times 25^2 = 100 \text{ m}$.

(b) Doppler at the bridge

(i) For a source moving towards a stationary observer:
 $f_{\text{obs}} = f \cdot v / (v - v_s)$.

Rearrange $\Rightarrow v_s = v - (f v) / f_{\text{obs}}$.

$$v_s = 340 - (270 \times 340) / 290 = 23.45 \text{ m s}^{-1} \approx 23 \text{ m s}^{-1}.$$

(ii) As the train passes and recedes, wavefronts are stretched at the observer, so the observed frequency decreases (Doppler effect).

Revision tips

- From rest with constant acceleration: $s = \frac{1}{2}at^2$.
- Doppler (source moving, observer stationary): $f_{\text{obs}} = f v / (v - v_s)$.
- Qualitative Doppler: bunching when approaching, stretching when receding.